ECET 250E-Linear Circuit 1
Problemset 7-Chapter 8- Second Order RLC Circuits

Name: _____ Marks: _____

Source Free Series RLC Circuits:

1. **Problem 8.13** For the circuit in Fig. 8.68, calculate the value of R needed to have a critically damped response.

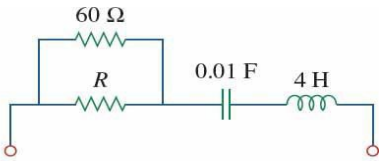


Figure 8.68
For Prob. 8.13.

2. **Problem 8.17** In the circuit of Fig. 8.71, the switch instantaneously moves from position A to B at $t = 0$. Find $v(t)$ for all $t \geq 0$.

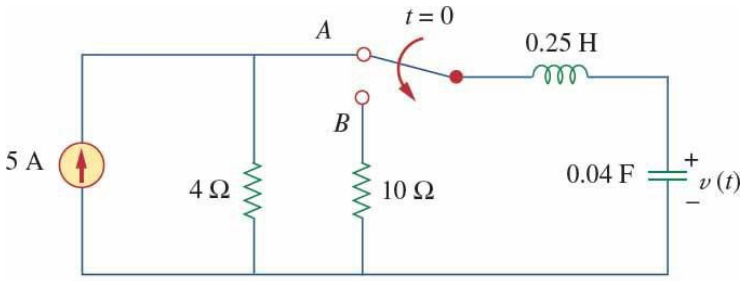


Figure 8.71
For Prob. 8.17.

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

SOURCE FREE PARALLEL RLC:

3. Problem 8.23 For the network in Fig. 8.76, what value of C is needed to make the response underdamped with unity damping factor ($\alpha = 1$)?

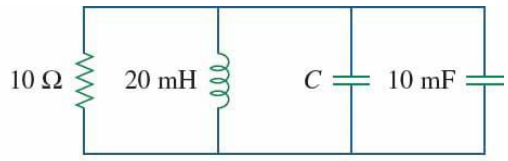


Figure 8.76
For Prob. 8.23.

Step response parallel RLC circuits:

4. **Problem 8.47** Find the output voltage $v_o(t)$ in the circuit of Fig. 8.94.

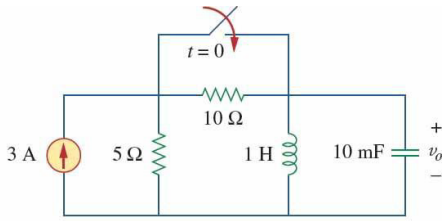


Figure 8.94
For Prob. 8.47.

5. **Problem 8.49** Determine $i(t)$ for $t > 0$ in the circuit of Fig. 8.96.

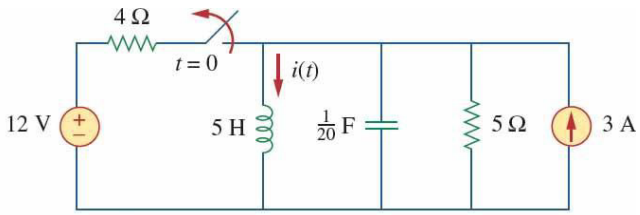


Figure 8.96
For Prob. 8.49.